

Assignment Report

ProcurePro Office Supplies Supplier Delay Risk Segmentation Unsupervised Learning for Pattern Discovery

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Task 1: Understand the Business Problem

What is the main business problem?

ProcurePro Office Supplies is a B2B procurement service that helps medium-sized organizations source, order, and manage office-related products. The main business problem is supplier delivery uncertainty. Some purchase orders arrive on time while others are delayed due to supplier backorders, low on-time performance, long delivery distance, urgent order pressure, seasonal demand spikes, or previous quality issues. ProcurePro wants to identify high-risk purchase orders before the promised delivery date so that procurement managers can take early action such as contacting suppliers, switching shipping methods, preparing backup suppliers, or alerting customers proactively.

Why is customer/user segmentation useful for this business?

Segmentation is useful because not all purchase orders carry the same level of delivery risk. By grouping purchase orders into segments based on their characteristics, ProcurePro can understand which types of orders are consistently high-risk and which are generally safe. This allows the business to allocate procurement resources more efficiently and design targeted risk management strategies for each group rather than treating all orders the same way.

What kind of marketing decisions could be improved by discovering customer/user groups?

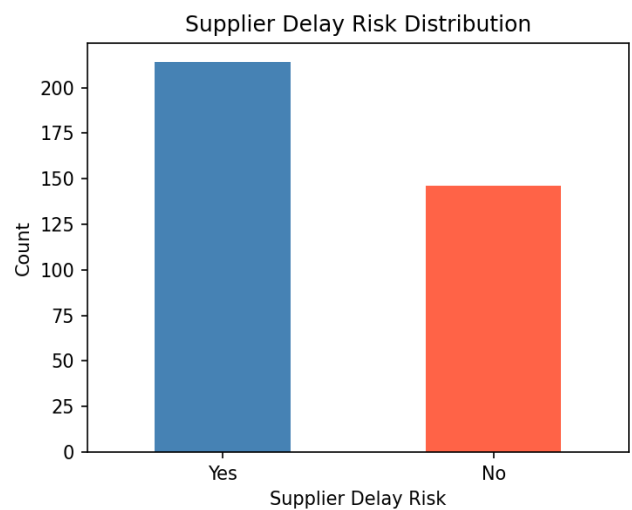
Discovering supplier and order groups can help ProcurePro make better procurement decisions such as which suppliers need closer monitoring, which orders should receive early reminders or backup supplier arrangements, which shipping modes are associated with delays, and which customer categories consistently face delivery problems. These insights can improve supplier relationship management, contract negotiations, inventory planning strategies, and customer communication.

Task 2: Prepare the Dataset

Dataset Overview

Item	Value
Total Rows	360
Total Columns	20
Missing Values	127
Duplicate Rows	0
Rows after Cleaning	360
Supplier Delay Risk: Yes	High Risk Orders Present
Supplier Delay Risk: No	Low Risk Orders Present

Target Variable Distribution



Selected Numerical Features for Clustering

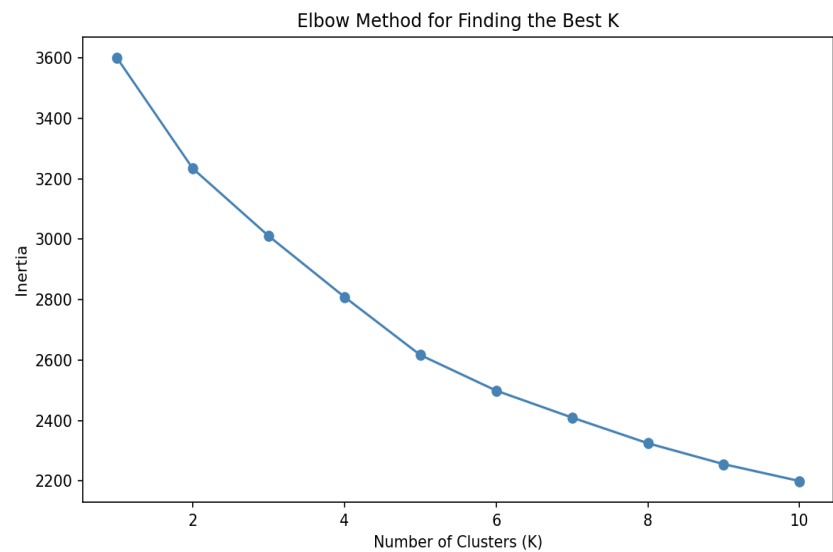
Feature	Business Meaning
Order_Value_CAD	Total value of the purchase order in Canadian dollars
Number_of_Line_Items	Number of product lines in the order
Promised_Lead_Time_Days	Number of days promised for delivery
Past_On_Time_Rate	Historical on-time delivery rate of the supplier
Supplier_Rating	Overall supplier performance rating
Quality_Incidents_Last_6M	Number of quality issues in the last 6 months
Prior_Delays_Last_6M	Number of delivery delays in the last 6 months
Distance_KM	Distance between supplier and delivery location
Inventory_Buffer_Days	Number of buffer days held in inventory
Seasonal_Demand_Index	Index reflecting seasonal demand pressure

Cleaning Steps

Missing numerical values were filled using the median of each column. Missing categorical values were filled using the mode. Purchase_Order_ID was dropped as it is only an identifier and not a useful feature for clustering. All selected numerical features were scaled using StandardScaler before applying K-means clustering to ensure no single feature dominates the distance calculations.

Task 3: Apply K-means Clustering

Elbow Method



The Elbow Method graph shows how inertia decreases as the number of clusters increases. From K=1 to K=3 the inertia drops sharply, meaning adding clusters significantly improves the grouping of purchase orders. After K=3 the decrease becomes much slower. The elbow point at K=3 indicates that 3 clusters is the most suitable choice for this dataset.

Why K=3 was Selected

K=3 was selected because the Elbow Method shows a clear bend at K=3 where the rate of inertia reduction slows down significantly. Three clusters also align well with natural business categories: high-risk orders that need immediate attention, moderate-risk orders that need monitoring, and low-risk orders from reliable suppliers. This makes the results practical and easy for procurement managers to interpret and act on.

Cluster Distribution

Segment	Number of Orders
Low Risk Reliable Suppliers	139
Moderate Risk Suppliers	125
High Delay Risk Suppliers	96

Task 4: Analyze and Interpret the Clusters

Cluster Summary

Feature	High Delay Risk	Moderate Risk	Low Risk Reliable
Order_Value_CAD	5,345.34	5,929.13	6,193.21
Past_On_Time_Rate	0.83	0.71	0.88
Supplier_Rating	3.99	3.15	4.22
Quality_Incidents_Last_6M	1.70	0.76	0.37
Prior_Delays_Last_6M	1.67	1.10	1.62
Distance_KM	552.17	627.02	1112.54
Inventory Buffer Days	8.96	17.44	17.65
Number of Orders	96	125	139

Segment 1: High Delay Risk Suppliers

This segment contains 96 purchase orders from suppliers with the highest number of quality incidents in the last 6 months at an average of 1.70 incidents. These suppliers have a moderate past on-time rate of 0.83 but the lowest supplier rating of 3.99 among all groups. The inventory buffer days are the lowest at 8.96, meaning there is very little safety stock to absorb delays. These orders require immediate procurement attention.

Segment 2: Moderate Risk Suppliers

This segment contains 125 purchase orders with the lowest past on-time rate at 0.71 and the lowest supplier rating at 3.15. These suppliers have fewer quality incidents at 0.76 but are located at an average distance of 627 km. Inventory buffer days are higher at 17.44, providing some protection against delays. These orders need close monitoring and proactive communication.

Segment 3: Low Risk Reliable Suppliers

This segment contains 139 purchase orders from the highest-rated suppliers with an average rating of 4.22 and the highest past on-time rate of 0.88. Quality incidents are very low at 0.37 per order. Despite being located the furthest away at an average of 1112 km, these suppliers consistently deliver on time. These orders are the safest and represent the benchmark for supplier performance at ProcurePro.

Most Valuable Segment

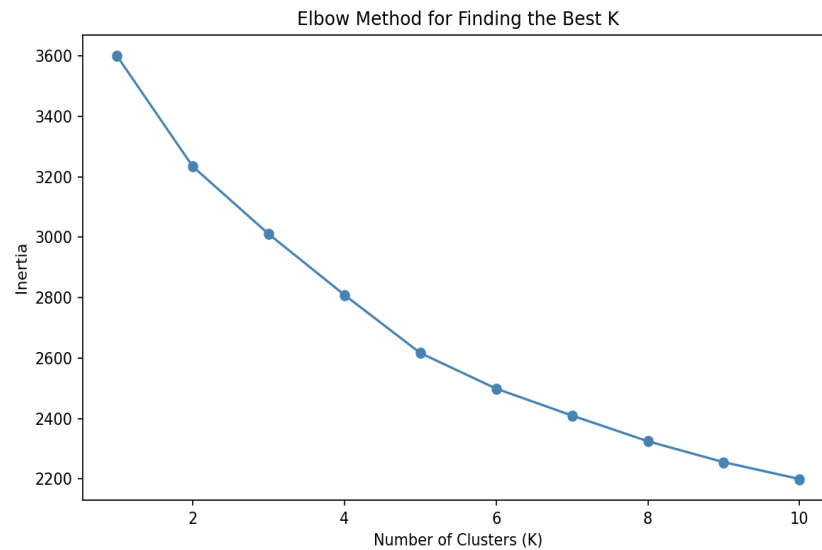
The Low Risk Reliable Suppliers segment is the most valuable for ProcurePro because these suppliers consistently deliver on time with high quality and strong ratings. Maintaining and strengthening these supplier relationships protects the business from delivery disruptions and reduces procurement management costs.

Segment Needing Most Marketing Attention

The High Delay Risk Suppliers segment needs the most attention because these orders have the highest quality incidents and lowest inventory buffer days. Without early intervention, these orders are most likely to cause delivery failures, customer dissatisfaction, and emergency procurement costs.

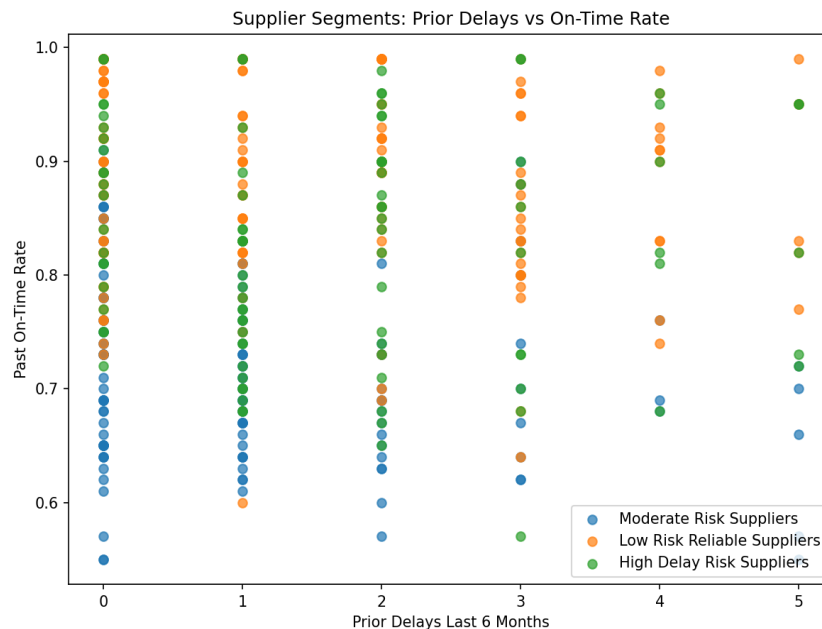
Task 5: Visualize the Clusters

Visualization 1: Elbow Method



The Elbow Method graph confirms that K=3 is the optimal number of clusters. The sharp drop in inertia from K=1 to K=3 shows that adding clusters up to this point significantly improves how well the data is grouped. After K=3 the improvement becomes marginal, making 3 clusters the most efficient and interpretable choice for this business problem.

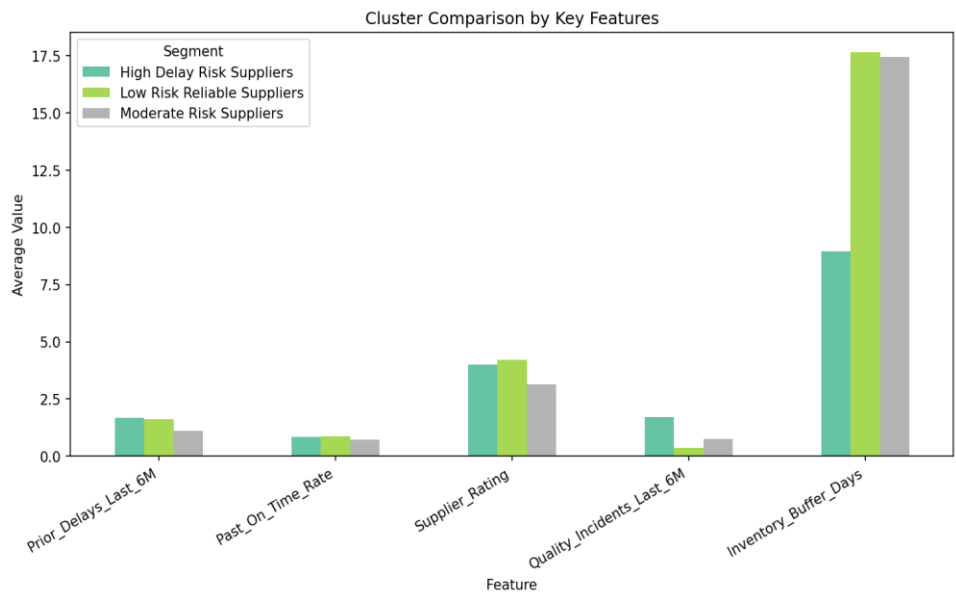
Visualization 2: Scatter Plot - Prior Delays vs On-Time Rate



This scatter plot shows how the three supplier segments are distributed based on Prior Delays Last 6 Months and Past On-Time Rate. High Delay Risk Suppliers appear with higher delays and lower on-time rates. Low Risk Reliable

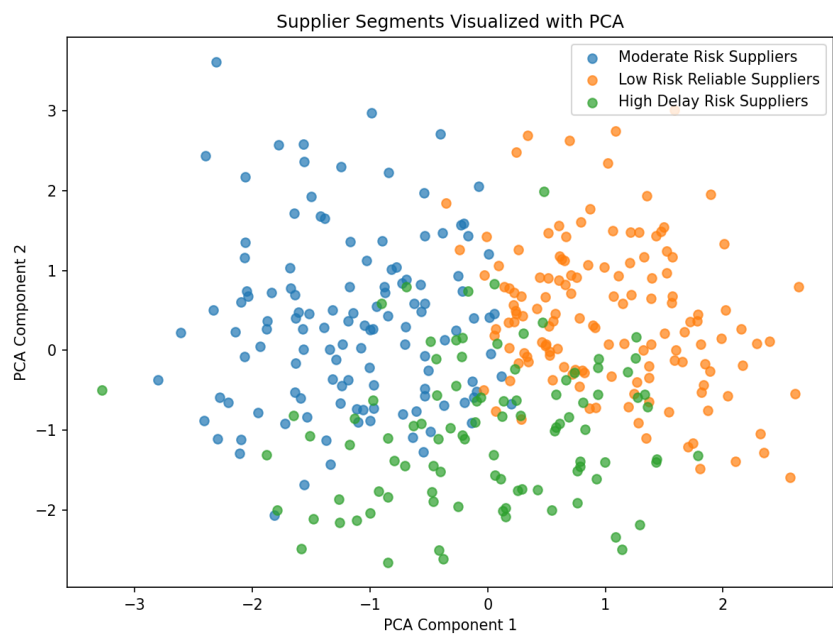
Suppliers show fewer delays and higher on-time rates. Moderate Risk Suppliers fall between the two groups. The clear separation between segments confirms that K-means identified meaningful differences in supplier delivery behaviour.

Visualization 3: Bar Chart - Cluster Comparison by Key Features



This bar chart compares the three supplier segments across five key features. High Delay Risk Suppliers have the highest quality incidents and prior delays. Low Risk Reliable Suppliers have the highest supplier ratings and on-time rates with the lowest quality incidents. Moderate Risk Suppliers fall between the two groups across all features. These clear differences confirm that each segment has a distinct risk profile that ProcurePro can use to design targeted procurement strategies.

Visualization 4: PCA Visualization



This PCA visualization reduces all ten features into two components to display the supplier segments in a two-dimensional space. PCA Component 1 explains 14.32% of variance and PCA Component 2 explains 13.20%, with a total of 27.52% explained variance. The three segments show visible separation in the PCA space, confirming that the K-means model identified patterns across multiple features simultaneously. The relatively low explained variance suggests that the ten features each contribute independently to cluster formation.

Task 6: Business Interpretation

What patterns did you discover?

Three distinct supplier segments were discovered based on purchase order characteristics and supplier performance history. The High Delay Risk segment is characterized by frequent quality incidents, low inventory buffers, and moderate on-time rates. The Moderate Risk segment has the lowest on-time rates and supplier ratings among all groups. The Low Risk Reliable segment shows consistently strong performance with high ratings, high on-time rates, and very few quality incidents.

How can the business use these segments?

ProcurePro can use these segments to prioritize procurement resources. High Delay Risk orders should receive immediate attention with backup supplier arrangements, increased inventory buffers, and early communication before order deadlines. Moderate Risk orders should be monitored closely with proactive reminders sent at key delivery milestones. Low Risk Reliable orders can be managed with standard processes while focusing energy on improving supplier relationships in the other two segments.

Marketing Strategy for Each Segment

Segment	Recommended Strategy	Business Goal
High Delay Risk Suppliers	Assign backup suppliers, increase inventory buffer, send early reminders, require frequent status updates	Prevent delivery failures and protect customer satisfaction
Moderate Risk Suppliers	Send proactive reminders, review shipping modes, offer performance improvement incentives	Move suppliers toward low-risk segment over time
Low Risk Reliable Suppliers	Offer preferred contract terms, volume commitments, and recognition programs	Maintain relationship and use as benchmark for other suppliers

How can this analysis help the business improve decision-making?

This analysis helps ProcurePro move from reactive to proactive procurement management. Instead of waiting for delays to happen, the business can use segment information to take early action on high-risk orders. It also helps managers allocate attention more efficiently by focusing on the suppliers and orders that need it most. Over time, tracking segment changes can show whether supplier performance is improving or declining, supporting better contract renewal decisions.

Task 7: Limitations and Responsible AI Reflection

What is one limitation of your dataset or clustering model?

The dataset is synthetic and does not reflect real-world procurement complexity. Factors such as weather disruptions, supplier strikes, sudden demand changes, customs delays, and new supplier improvements are not captured in the data. This means the clusters may not fully represent actual supplier behaviour. The PCA explained variance of only 27.52% also suggests that the ten features have complex relationships that are not fully captured in two dimensions.

Why does K-means not always produce perfect customer segments?

K-means requires the number of clusters to be chosen before training, which is not always obvious from the data. It assumes clusters are roughly circular and equally sized, which may not match real supplier patterns. K-means is also sensitive to outliers and may group unusual orders incorrectly. Different random starting points can produce slightly different results, and the algorithm treats all features as equally important even when some features may be more relevant than others.

What kind of bias or unfair decision could happen without human review?

The model may unfairly associate certain supplier regions or categories with high delay risk based on historical patterns that no longer apply. For example, a supplier from a region that previously had delays may have significantly improved but could still be flagged as high risk. Using the model without review could damage supplier relationships, lead to unfair contract decisions, result in missed opportunities with improving suppliers, or create unfair disadvantages for suppliers from certain geographic areas.

Why should human judgment still be used when making marketing decisions?

The model is a decision-support tool, not a replacement for human expertise. Procurement managers understand supplier relationships, current market conditions, contract obligations, and customer priorities that are not captured in the data. Human review ensures that model predictions are interpreted in context and that suppliers are treated fairly. A responsible procurement workflow would use the model to flag orders for review, but employees should decide the best action based on supplier communication history, current inventory status, contract terms, and customer importance.

Business Analysis

SWOT Analysis

The SWOT analysis below evaluates the strengths, weaknesses, opportunities, and threats of using K-means clustering for ProcurePro supplier delay risk segmentation.

Strengths	Weaknesses
- Simple and explainable segments procurement teams can act on - Automates risk flagging across hundreds of purchase orders - Prior_Delays_Last_6M and Past_On_Time_Rate are strong clustering signals - Reduces manual review workload for procurement managers - Helps design targeted strategies for each supplier group	- K-means requires number of clusters to be chosen manually - PCA explains only 27.52% of total variance - Synthetic dataset may not reflect real procurement complexity - Model is sensitive to outliers in order data - Segment boundaries may shift as new order data arrives

Opportunities	Threats
- Integrate clustering into ProcurePro procurement dashboard - Add real-world data such as weather and supplier financial health - Use segments to trigger automated supplier notifications - Expand model to predict expected delay duration not just risk - Use Low Risk segment as benchmark for supplier development	- Unfair association of certain regions with high delay risk - Over-reliance on model without human review may damage supplier relationships - Historical data may not reflect current supplier improvements - Model may become outdated as supply chain conditions change - Misclassification of improving suppliers as high risk

Model Analysis

K-means clustering was selected for this problem because it is an unsupervised learning method that discovers hidden patterns in purchase order data without requiring a predefined target variable. The model was applied to 360 purchase orders using 10 numerical features after StandardScaler normalization.

Aspect	Detail
Model Type	K-means Clustering
Number of Clusters	3 (selected using Elbow Method)
Total Records	360 purchase orders
Features Used	10 numerical features
Scaling Method	StandardScaler
PCA Explained Variance	27.52% (PCA1: 14.32%, PCA2: 13.20%)
Segment 1	Low Risk Reliable Suppliers (139 orders)
Segment 2	Moderate Risk Suppliers (125 orders)
Segment 3	High Delay Risk Suppliers (96 orders)
Most Important Features	Prior_Delays_Last_6M, Past_On_Time_Rate, Quality_Incidents_Last_6M

Recommendations

Based on the clustering results and business analysis, the following recommendations are made for ProcurePro to improve supplier delay risk management.

Recommendation	Business Impact
Flag all High Delay Risk orders automatically and assign dedicated procurement managers.	Prevents delivery failures and reduces emergency procurement costs.
Increase inventory buffer days for High Delay Risk segment orders.	Provides safety stock to absorb potential delays without affecting customers.

Send automated proactive reminders to Moderate Risk suppliers at key delivery milestones.	Reduces the chance of delays escalating and moves suppliers toward the low-risk segment.
Offer preferred contract terms and volume incentives to Low Risk Reliable suppliers.	Strengthens relationships with the best performing suppliers and encourages consistency.
Retrain the clustering model quarterly as new purchase order data becomes available.	Keeps segments accurate and aligned with current supplier performance trends.
Always combine model output with supplier communication before taking contract action.	Ensures fair treatment of suppliers and prevents decisions based on outdated data.

Overall Recommendation

The K-means clustering model provides ProcurePro with a practical framework for understanding supplier delivery risk across three distinct segments. The High Delay Risk segment with 96 orders requires immediate attention due to high quality incidents and low inventory buffers. The Moderate Risk segment with 125 orders needs structured monitoring and improvement programs. The Low Risk Reliable segment with 139 orders should be strengthened through preferred partnerships and used as the performance benchmark for the rest of the supplier base. Combined with human oversight, regular model retraining, and proactive supplier communication, this segmentation approach will help ProcurePro reduce delivery failures, lower emergency procurement costs, and build a more resilient and reliable supply chain.